

Ideation Phase

Empathize & Discover

Date	24 June 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	4 Marks

Empathy Map Canvas

User Persona

Primary User: Farmer

The Smart Agricultural Production Optimization Engine (OptiCrop) is primarily designed for farmers who need reliable, data-driven support in selecting suitable crops based on soil nutrients and environmental conditions. The empathy map was developed to better understand the farmer's goals, challenges, behaviors, and expectations before designing the proposed solution.

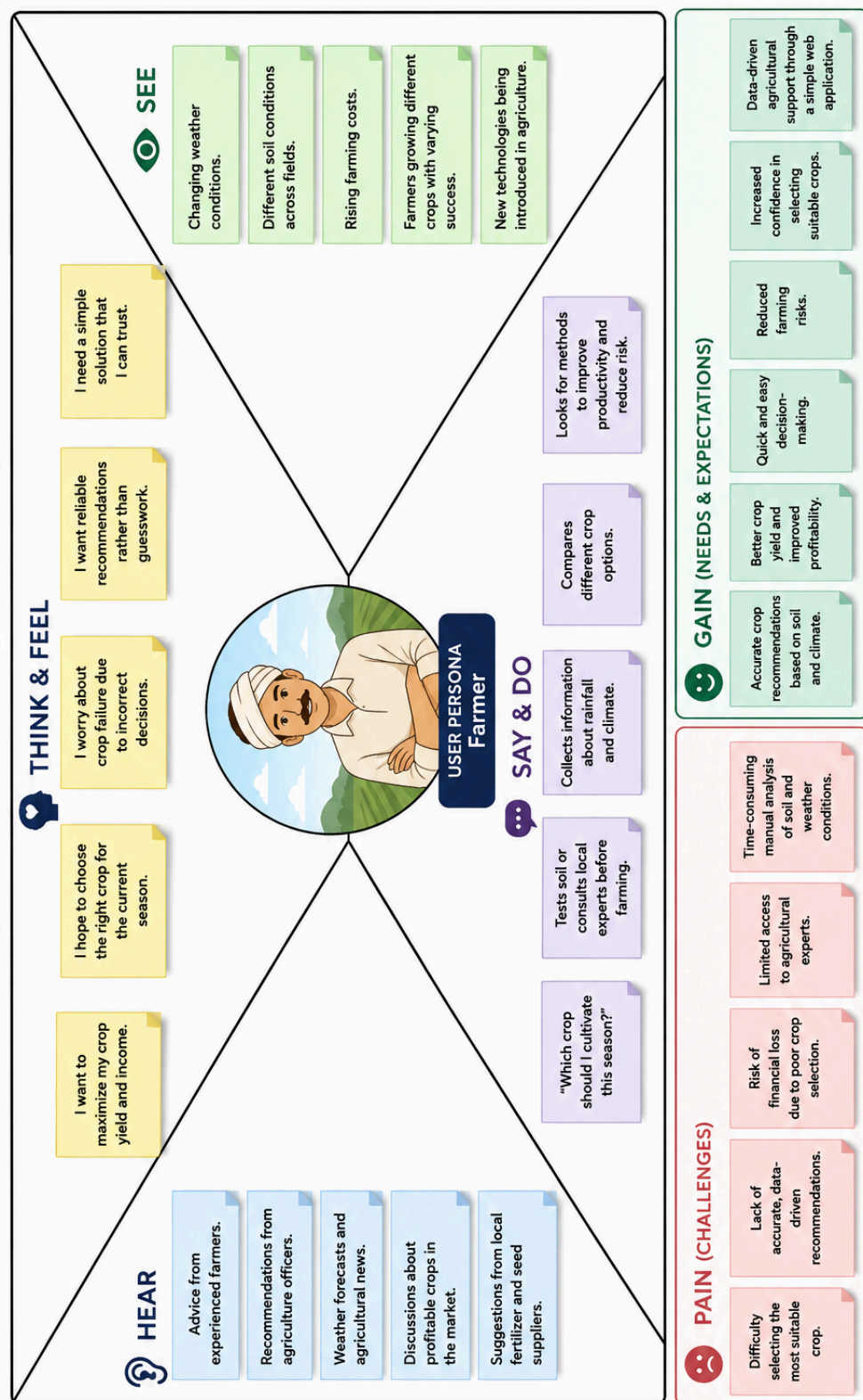
Empathy Map Description

Think & Feel

- Wants to maximize crop yield and income.
- Hopes to choose the right crop for the current season.

EMPATHY MAP CANVAS

Project: Smart Agricultural Production Optimization Engine (OptiCrop)



- Worries about crop failure due to incorrect decisions.

- Prefers reliable recommendations instead of guesswork.
- Needs a simple and trustworthy decision-support system.

See

- Changing weather conditions.
- Different soil conditions across agricultural fields.
- Rising farming costs.
- Farmers cultivating different crops with varying success.
- Increasing adoption of modern agricultural technologies.

Hear

- Advice from experienced farmers.
- Recommendations from agriculture officers.
- Weather forecasts and agricultural news.
- Discussions about profitable crops in the market.
- Suggestions from local fertilizer and seed suppliers.

Say & Do

- "Which crop should I cultivate this season?"
- Tests soil or consults local agricultural experts.
- Collects rainfall and climate information.
- Compares different crop options before cultivation.
- Looks for methods to improve productivity and reduce farming risks.

Pain (Challenges)

- Difficulty selecting the most suitable crop.
- Lack of accurate, data-driven recommendations.
- Risk of financial loss due to poor crop selection.
- Limited access to agricultural experts.
- Time-consuming manual analysis of soil and weather conditions.

Gain (Needs & Expectations)

- Accurate crop recommendations based on soil and climate.
- Better crop yield and improved profitability.
- Quick and easy decision-making.
- Reduced farming risks.
- Increased confidence in crop selection.

- A simple web-based application that provides reliable recommendations using machine learning.

Conclusion

The empathy mapping exercise helped the team understand the real needs and challenges faced by farmers. These insights guided the development of the Smart Agricultural Production Optimization Engine (OptiCrop), ensuring that the system focuses on providing accurate crop recommendations, reducing farming risks, and improving agricultural productivity through machine learning.

